

Sample Project Documentation

Field	Details
Starting Date	08 October 2026
Team ID	UNM1611PUNITHA
Project Name	AI STUDY BUDDY API

Project Overview

AI Study Buddy API is a backend service for students to manage study materials and use AI-assisted learning operations. The supplied implementation uses Node.js, Express, MongoDB/Mongoose, JWT authentication, Multer uploads and Google Gemini integration.

Functional Scope

Module	Implemented capability
Authentication	Student registration, login, refresh token and logout endpoints
Authorization	JWT protection and Student/Admin role checks
Study materials	Upload, list, get and delete material operations
AI assistance	Summarization, flashcards, MCQ quiz and personalized study-plan generation
Administration	List users, delete users and view admin statistics
Security helpers	bcryptjs password hashing, JWT tokens, upload filtering and size limit

API Endpoint Reference

Method	Route	Purpose	Auth
POST	/register	Create account	Public
POST	/login	Authenticate user	Public
POST	/refresh	Refresh access token	Refresh token
POST	/logout	Logout response	Authenticated
POST	/materials/upload	Upload study material	Authenticated
GET	/materials/	List materials	Authenticated
GET	/materials/:id	Get one material	Authenticated
DELETE	/materials/:id	Delete owned material	Authenticated
POST	/materials/:id/summarize	Generate summary	Authenticated
POST	/materials/:id/flashcards	Generate flashcards	Authenticated
POST	/materials/:id/quiz	Generate MCQ quiz	Authenticated
POST	/materials/:id/study-plan	Generate study plan	Authenticated
GET	/admin/users	List users	Admin
DELETE	/admin/users/:id	Delete user	Admin
GET	/admin/stats	View statistics	Admin

Architecture & Data Flow

Client/API request → Express route → authentication/authorization middleware where required → controller → MongoDB/Mongoose and/or Gemini/file-processing utilities → JSON response.

Component	Responsibility
Express	HTTP routing and middleware pipeline
JWT middleware	Authenticates bearer access tokens and applies role checks
Mongoose	Persists User and Material documents in MongoDB
Multer	Processes uploaded study-material files and applies upload constraints
Gemini utility	Sends prompts to Google Gemini and returns generated content
Controllers	Coordinate validation, persistence, file handling and AI operations

Authentication

The source implementation uses JWT access and refresh tokens. Requests to protected endpoints use the access token through the Authorization Bearer mechanism, with refresh-token handling implemented separately. Passwords are hashed with bcryptjs before persistence.

Database Models

Model	Purpose
User	Stores account identity, authentication-related data and role information
Material	Stores uploaded study-material metadata/content and ownership information

File Upload Rules

The Multer middleware allows .txt, .md and .pdf extensions and applies a 5 MB file-size limit. Uploaded files are processed from temporary storage. The supplied source should not be described as providing full PDF text extraction because the controller reads the uploaded file using UTF-8 text handling rather than a dedicated PDF parser.

AI Features

Feature	Operation
Summary	Generates a concise summary from study material
Flashcards	Generates flashcard content using Gemini and parses the returned JSON
Quiz	Generates MCQ quiz content using Gemini and parses the returned JSON
Study plan	Generates a personalized study plan using Gemini

Security & Error Handling

The project uses password hashing, JWT authentication, role-based authorization, ownership checks for materials, upload type/size validation and centralized asynchronous error handling. Sensitive environment configuration should remain outside source control.

Known Limitations

Limitation	Documentation note
Automated tests	No implemented test suite is supplied; package.json uses a placeholder test script.
Run command mismatch	package.json references server.js while the supplied source listing contains index.js.
PDF handling	PDF extension is accepted, but no dedicated PDF parser is present in the supplied dependencies/source.
Logout semantics	Logout does not show server-side token revocation in the supplied implementation.
AI response validation	Generated JSON is parsed directly without an explicit schema validation layer.

Submission / Evaluation Fields

Marks / evaluation allocation should be completed by the evaluator using the original GitHub assessment scheme. No score has been fabricated in this document. The supplied project evidence, limitations and execution status are stated explicitly so that marks can be awarded against actual deliverables.

Documentation criterion	Evidence included	Marks / Score
Project overview & scope	Overview and functional scope	Evaluator entry
Architecture & API documentation	Architecture, endpoints and components	Evaluator entry
Setup / executable instructions	Run instructions and configuration guidance	Evaluator entry
Limitations / accuracy	Known implementation limitations recorded	Evaluator entry